IRENA
International Renewable Energy Agency

RENEWABLE POWER GENERATION COSTS IN 2024

ISBN: 978-92-9260-669-5

Citation: IRENA (2025), *Renewable power generation costs in 2024*, International Renewable Energy Agency, Abu Dhabi.

About IRENA

The International Renewable Energy Agency (IRENA) is an intergovernmental organisation that supports countries in their transition to a sustainable energy future and serves as the principal platform for international co-operation, a centre of excellence, and a repository of policy, technology, resource and financial knowledge on renewable energy. IRENA promotes the widespread adoption and sustainable use of all forms of renewable energy, including bioenergy, geothermal, hydropower, ocean, solar and wind energy, in the pursuit of sustainable development, energy access, energy security and low-carbon economic growth and prosperity. **www.irena.org**

Acknowledgements

This report was authored by Saied Dardour, Deborah Ayres and Lourdes Zamora (IRENA), under the direction of Norela Constantinescu (Acting Director, IRENA Innovation and Technology Center).

The authors are grateful for the valuable contributions provided by IRENA colleagues, including Adrian Gonzalez, Binu Parthan, Diala Hawila, Faran Rana, Erick Ruiz Araya, Julian Prime, Simon Benmarraze, Yasuhiro Sakuma, Aakarshan Vaid (Consultant), Alina Gilmanova (Consultant) and Olga Sorokina (European Energy Link Group) in the preparation of this study.

This report benefited from the reviews and comments of numerous experts, including: Ana Andrade (Direção Geral de Energia e Geologia), Ana Rovzar (Global Renewables Alliance / GWEC), Anjali Lathigara (Global Wind Energy Council), Bruce Douglas (Global Renewables Alliance), Christian Kirchsteiger (European Commission), Christoph Wolter (Danish Energy Agency), Fabien Perdu (Ministry of Economy, Trade and Industry, Japan), Feng Zhao (Global Wind Energy Council), Guy Brindley (WindEurope), Hal Turton (KAPSARC), Jarett Zuboy (National Renewable Energy Laboratory), Joyce Lee (Global Wind Energy Council), Manuel Quero (Sunntics), Marco Cometto (IAEA), Michael Taylor (ex-IRENA), Nouha Gazbour (CEA France), Rebecca Ellis (International Hydropower Association), Rebecca Williams (Global Wind Energy Council), Stéphane Cathalau (CEA France), Yuetao Xie (China Renewable Energy Engineering Institute), and Zhou Xiaoxi (China Renewable Energy Engineering Institute). All opinions and errors remain those of the authors.

Publications, editorial and communications support were provided by Francis Field, Stephanie Clarke, Nicole Bockstaller and Daria Gazzola.

The report was copy-edited by Jonathan Gorvett, and a technical review was provided by Paul Komor. Graphic design was provided by Nacho Sanz.

For further information or to provide feedback: **publications@irena.org**
This report is available for download: **www.irena.org/publications**

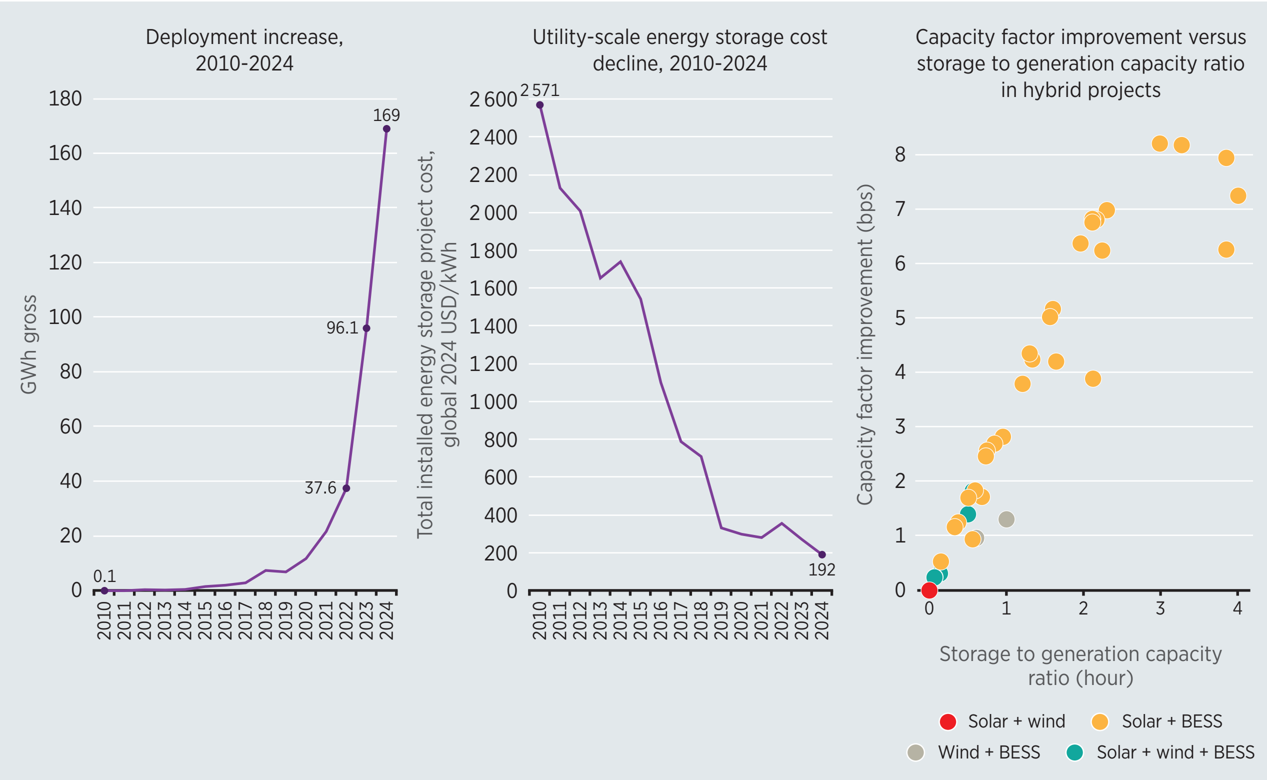
09 ENABLING TECHNOLOGIES

Aya Photo © Shutterstock.com

HIGHLIGHTS

- Between 2010 and 2024, the costs of battery storage projects declined 93%, from USD 2 571/kWh to USD 192/kWh. This cost reduction was driven by technological advancements in battery chemistries, materials efficiency improvements, manufacturing scale-up and optimisation, as well as increased market competition.
- Annual capacity additions for chemical battery storage increased from 0.1 GWh gross capacity in 2010 to 169 GWh gross capacity in 2024.
- China led in terms of new additions, installing 84 GWh in 2024 and accounting for half of total global additions.
- The United States was the second largest market, adding 41 GWh of BESS and representing almost a quarter of the total new added capacity.
- Energy shifting is the main application of electricity storage, accounting for 68% of the total added capacity in 2024. Energy shifting balances the power system by storing renewable energy production at times of low market prices (leading to low/null revenue), or low demand (leading to curtailments).
- The commissioning of renewable energy projects coupled with of battery energy storage systems (BESS) has increased over the past two years, and more of these projects are expected to come online in the near future, ensuring greater grid flexibility.
- In hybrid wind and solar projects, increasing the storage-to-generation capacity ratio generally leads to greater improvements in capacity factor. However, these gains must be balanced against the upfront costs of BESS.

Figure 9.1 Global deployment increase and utility-scale energy cost decline, 2010–2024 (left) and capacity factor improvement in hybrid systems (right)



Notes: bps = basis points; BESS = battery energy storage system; GWh = gigawatt hour; kWh = kilowatt hour; USD = United States dollar.

INTRODUCTION

The energy transition requires urgent and substantial acceleration across the energy supply and end-use sectors, as well as the leveraging of technologies to achieve global climate goals. Currently, the rising competitiveness of renewables in comparison to fossil fuels is driving extensive deployment of the former, and the goal to triple renewable energy capacity by 2030 agreed at COP28 sets an ambitious and significant target. In the next few years of this decade, extensive deployment of renewable power generation in more countries and regions is expected. As the electricity system evolves, enabling technologies are also being introduced to facilitate the transition from a fossil fuel-based system to one based on renewables (IRENA, 2024).

Today, a wide variety of energy storage solutions are available, including: electrochemical storage (notably lithium-ion batteries, but also flow batteries and those using other chemistries); thermal energy storage (which employs rocks, bricks or molten salts to store heat); mechanical technologies (using compressed air, liquid air or gravitational potential such as in conventional and novel pumped hydro); and chemical storage (storing energy in chemical bonds, such as hydrogen or its derivatives).

With a growing share of variable renewable power generation, energy storage will play a crucial role in ensuring the successful delivery of a reliable electricity system. By capturing surplus electricity and releasing it when needed, batteries enhance system flexibility, cut transmission losses and relieve grid congestion. Energy storage technologies, such as chemical batteries or pumped hydro storage, have fast-response capabilities and can stabilise frequency, bolster resilience during outages and provide backup power, thereby safeguarding grid security. At the same time, storage underpins the clean energy transition by reducing dependence on fossil-fuels and lowering overall emissions. Finally, by deferring costly transmission and distribution upgrades, and enabling market arbitrage opportunities, battery systems deliver significant economic value to both grid operators and end users.

Electricity storage is already an important tool in minimising overall electricity system costs – mainly using pumped hydropower storage systems. It can also provide ancillary services to the electricity market and offer energy arbitrage. Ancillary services provide stability and reliability to the grid, while energy arbitrage often involves storing electricity at times of relative surplus, such as overnight, for release when wholesale prices or generating costs are higher. However, it is noteworthy that electricity flexibility needs encompass time scales that range from seconds to entire seasons. Electricity storage, particularly in the form of batteries, is entirely relevant, but only up to daily time scales. It cannot solve, for example, the mismatch between PV production in summer and maximum energy demand in winter for space heating in many northern countries. Unlike short-duration batteries, long-duration energy storage (LDES) can support a broader range of grid services — from supplying short-term reserves to addressing extended energy shortfalls. It also helps minimise reliance on coal and gas power plants, enables continuous use of renewable energy, and can delay the need for major grid infrastructure upgrades (BNEF, 2021).

Storage can be coupled directly with solar PV or wind (as with hybrid generation assets), or function as a standalone asset, optimising distribution and transmission systems. Other hybrid system configurations exist and a growing number of hydropower and pumped hydro storage projects are being co-located with wind, ground-mounted solar PV systems as well as floating solar installations. Hybrid systems deliver more predictable, dispatchable power profiles, helping to mitigate the impacts of grid connection delays, curtailment risks, and network congestion, by absorbing excess generation and smoothing demand peaks. As grid modernisation efforts advance, co-locating storage with renewables is proving to be a key enabler for accelerating renewable deployment. Battery storage is increasingly being deployed in hybrid configurations, with renewable technologies, enhancing their value and grid compatibility.

LONG DURATION ENERGY STORAGE

As the energy transition accelerates to 2030, the system will need long-duration energy storage (LDES), as well as a combination of other flexibility measures (bioenergy for power, geothermal, reservoir storage hydropower, demand-side management, interconnectors, *etc.*). LDES technologies are well adapted to ensure the resilience of the electricity system thanks to their capacity to discharge over long periods of time, especially in a system relying on more variable renewable sources. LDES offers durations of at least six hours and can complement short duration applications, such as lithium-ion batteries.

A variety of LDES technologies are emerging, including chemical, electro-chemical, thermal and mechanical. The global total installed capacity of long duration energy storage reached 1 GW in 2024, representing an energy capacity of 4.6 GWh (Zhou, 2024). Currently, the electricity storage landscape is dominated by conventional pumped hydroelectricity storage (PHS), with around 150 GW of cumulative power capacity in 2024 (IRENA, 2025d). While first used to support nuclear fleets, many countries are expanding and increasingly utilising their pumped storage hydropower capacity to enhance electricity storage and grid flexibility for variable renewable integration. Pumped storage hydropower (PSH), is increasingly utilised in Europe (Germany, Switzerland and Spain) and in Asia (China and India), expanding national capacities to enhance grid flexibility (Ember, 2025a; GLOBSEC, 2024).

While LDES technologies often remain costlier than lithium-ion batteries, continued innovation and supportive policies are key to reducing costs and unlocking broader deployment potential. Costs depend on the technology, project size, storage duration and location. Conventional pumped storage is still the most competitive technology, with a global average installed cost of USD 156/kWh; thermal storage, such as molten salt and solid state, has a global average installed cost of USD 238/kWh. For emerging technologies, costs range from USD 300/kWh for compressed air energy storage (CAES) to USD 658/kWh for gravity energy storage (BNEF, 2024a). The viability and performance of LDES for extended storage needs are expected to improve as technology advances and practical experience grows. Early adoption and faster commercialisation may also depend heavily on favourable regulatory and policy frameworks.

An increasing number of markets – particularly those with high renewable energy penetration – are expected to implement policies, market mechanisms and financial incentives to promote LDES deployment. Targets for long-duration energy storage have been established at the local level in Australia, Canada and China (Zhou, 2024).

UTILITY-SCALE BATTERY STORAGE DEPLOYMENT COST TRENDS

Stationary storage complements and facilitates the rapid rise of installed electricity generation capacity in solar PV and wind. By balancing supply and demand, as well as providing ancillary electricity market services, storage technologies enhance system stability and flexibility (IRENA, 2023). This strategic role is reflected in the sharp growth of global deployment in recent years.

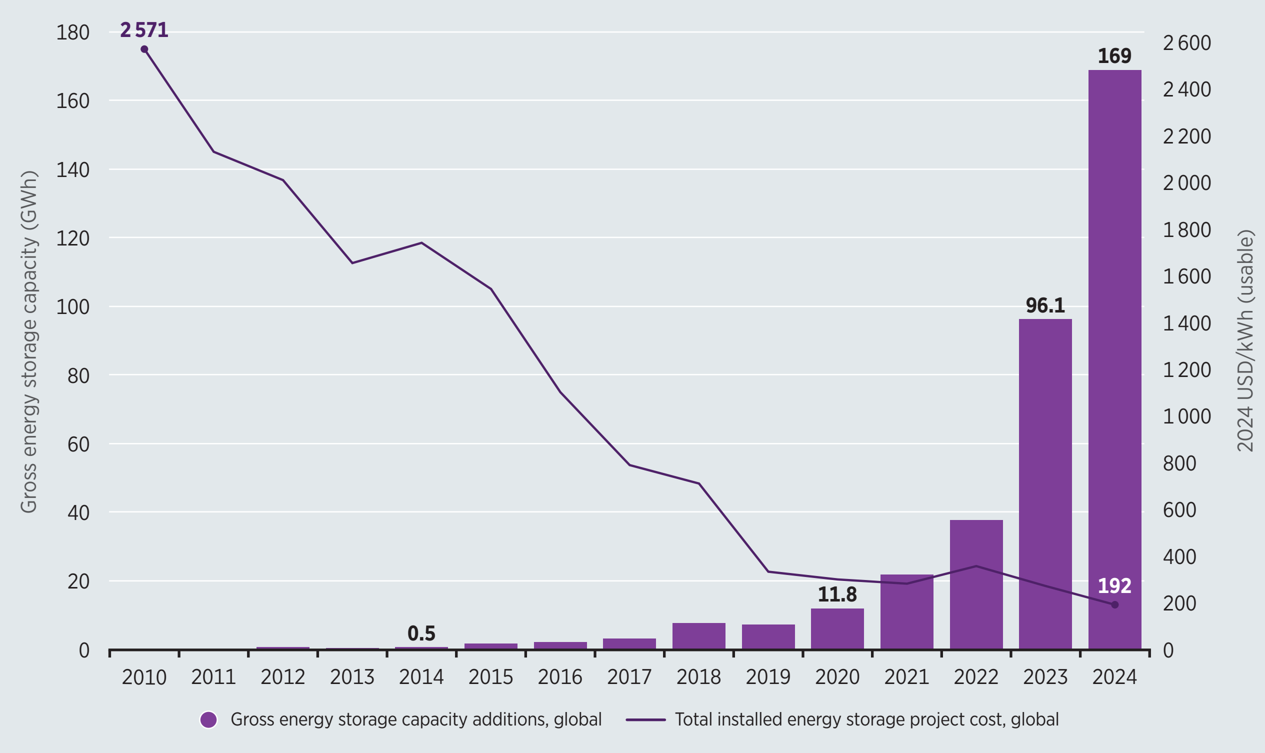
Between 2010 and 2024, new annual capacity additions of battery energy storage increased from 0.1 GWh of gross capacity in 2010 to 169 GWh of gross capacity in 2024 – a year that witnessed a record annual increase (Figure 6.2). China and United States remained the markets leaders in 2024. China installed the most capacity, at 84 GWh (36 GW) – representing an increase of 80% compared to the annual additions in 2023 and accounting for half of total global additions. The United States was the second largest market, adding 41 GWh (13 GW), which was almost a quarter of total new added capacity.

In China and the United States, the rapid increase in capacity was mainly driven by regional and local energy storage mandates and targets. These will continue to foster deployment in the period ahead. The commissioning of renewable energy projects coupled with battery energy storage systems (BESS) has been increasing over the past two years, and more are expected to come online in the coming years ensuring greater grid flexibility.

In other regions, such as Europe, the United Kingdom, Germany and Italy lead the annual utility-scale storage deployment, accounting for 62% of total capacity additions in 2024 (BNEF, 2024b). This includes a mix of pumped hydro storage, grid-scale batteries and behind-the-meter systems (Petrovich and Fox, 2024).

Meanwhile, in the Middle East and North Africa, countries such as Saudi Arabia, the United Arab Emirates and Egypt are leading investments in giga-scale energy storage projects; these recent additions are entirely electrochemical, reflecting a strong shift toward BESS (Aruffo and Matthes, 2025). Across both regions, local supply chain development has been a crucial strategy to reduce costs and enhance the resilience of energy storage deployment.

Figure 9.2 Global gross battery storage capacity additions by year and total installed electricity storage project costs per kWh, 2010–2024



Source: (BNEF, 2024; Schmidt and Staffell, 2023).

Notes: Cost data from 2010 to 2015 was calculated based on the capacity, price and experience curve regression data for electrical energy storage technologies model developed by Oliver Schmidt and Iain Staffell; GWh = gigawatt hour; kWh = kilowatt hour; USD = United States dollar.

Globally, the costs of a fully installed and commissioned battery storage project declined by 93% between 2010 and 2024, from USD 2 571/kWh to USD 192/kWh (Figure 9.2) (BNEF, 2024c). In recent years, this cost reduction has been driven by technological developments that have improved materials efficiency and manufacturing processes, leading to economies of scale. From a manufacturing perspective, the market has been growing, creating competition between suppliers across the entire value chain. Production has rapidly expanded, especially in China, where the supply chain had already established a robust scale of production that was capable of delivering in a short period of time. In 2024, abundant supply capacity and competition are the main drivers of cost declines in energy storage systems (BNEF, 2024d).

Figure 9.3 presents turnkey[48] battery storage system prices by market and duration between 2021 and 2024. Supply chain disruptions and volatility of raw material costs pushed up prices in 2022 but turnkey energy storage costs have fallen rapidly in recent years, reaching USD 165/kWh in 2024 – this value is 40% lower compared to 2023. One of the main drivers of the cost reduction is the increase of global manufacturing, leading first to technological improvements and higher performances in energy density and specific energy, and thereafter to economies of scale with increasing factory size and throughput.

[48] *Turnkey battery storage system costs include project equipment based on usable capacity, excluding EPC and grid connection costs*

Figure 9.3 Turnkey battery storage system prices by market and duration, 2021–2024

Based on: (BNEF, 2024d).
Notes: In the lower half of the chart, the reference line represents the cost ratio compared to the global average; kWh = kilowatt hour; USD = United States dollar.

Turnkey energy storage system costs are available for major markets, with these providing a sense of the cost variation between regions. These costs include the price of battery providers and system integrators and exclude EPC, grid connection and development costs. In 2024, global weighted average turnkey energy storage costs ranged between USD 165/kWh and USD 148/kWh, according to system duration. As the duration of the system increases from two to four hours, the cost decreases by 11%. Battery storage costs decreased 38% for a 2-hour system and 32% for a 4-hour system compared to 2023.

Battery cell prices, which are closely linked to the cost of key raw materials such as lithium, nickel and cobalt, remained relatively low in 2024. This trend was partly driven by the stabilisation or decline in battery metal prices as new mining and refining capacity came online, softening demand expectations (BNEF, 2024e). The future evolution of battery cell costs remains uncertain, as it will be dominated by these material costs, the volatility of which is increasing with demand.

China maintains the lowest battery storage costs due to its robust manufacturing scale and supply chain efficiencies, while Europe and the United States continue to experience higher costs due to import reliance (BNEF, 2023).

In China – the most competitive market – prices are around 40% less expensive than the global average. In 2024, electricity storage in China saw a cost decrease ranging from 48% (for a 4-hour system) to 47% (for a 2-hour system). Lower prices in China are mainly due to its well-established supply chain and large manufacturing capacity, which creates strong domestic market competition, maintaining downward pricing pressure that will endure in the years ahead.

Europe had the highest cost ratio compared to global prices, and the lowest cost decrease between 2023 and 2024, ranging from 16% (for a 2-hour system) to 20% (for a 4-hour system) for all storage systems. Battery economics have improved in European countries due to the growth of renewables. The United Kingdom is the biggest market in grid-scale BESS deployment, and also leads in hybrid solar PV plus BESS installations, which account for 62% of the total installed capacity in the country. This leadership is driven by strong policy support, favourable market conditions, and the development of some of the largest hybrid energy parks in Europe (SolarPower Europe, 2025).

In 2024, the United States registered a cost ratio compared to global prices of 1.50 for a 2-hour and 1.58 for a 4-hour battery system. Prices decreased 34% and 27%, respectively, compared to 2023, and developers willing to pay more for products made in the United States have become more common in recent years, driven by efforts to diversify away from Chinese supply chains to avoid policy uncertainties such as the threat of rising tariffs, or in response to utility requirements to avoid Chinese suppliers (BNEF, 2024f).

UTILITY-SCALE BATTERY STORAGE TECHNOLOGY MARKET SHARE

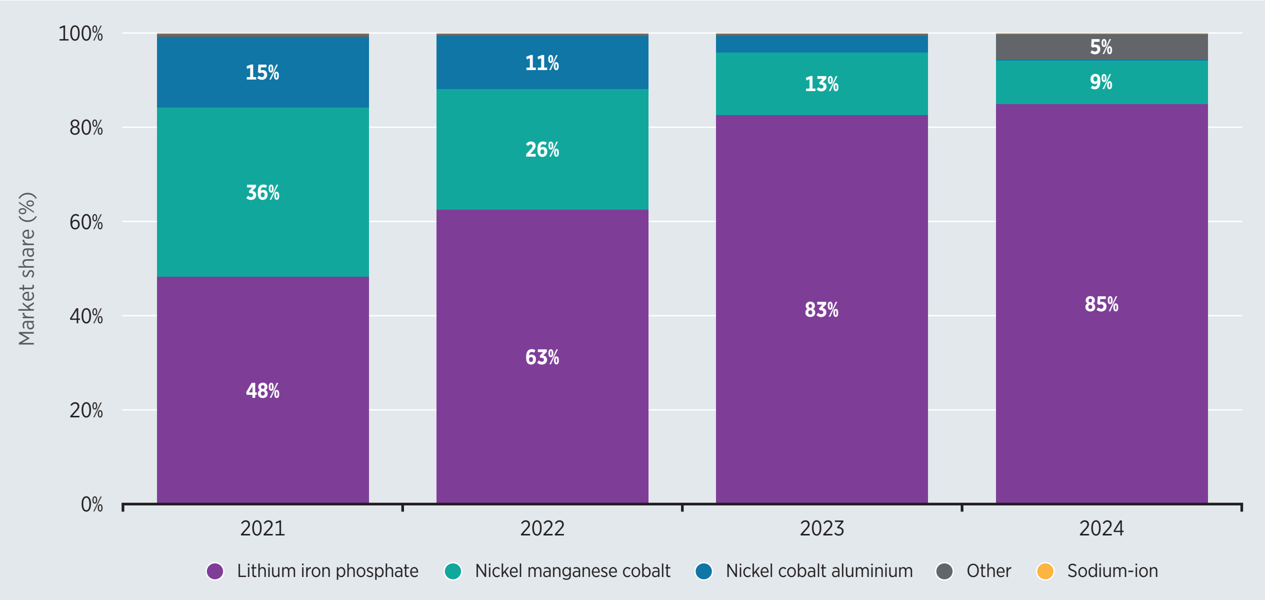
A wide range of battery technologies has emerged to meet various performance, cost and sustainability requirements. Key chemistries include: lithium iron phosphate (LFP), nickel manganese cobalt (NMC), nickel cobalt aluminium (NCA), and sodium-ion (Na-ion) batteries, among others.

Figure 9.4 illustrates the evolving market share of utility-scale battery storage technologies from 2021 to 2024. Overall, lithium-ion batteries have dominated the market due to their high energy density, efficiency, proven scalability and lifecycle advantages (IEA, 2024).

A clear trend toward the increasing dominance of LFP batteries is seen. LFP batteries grew from 48% in 2021 to an estimated 85% by 2024, thanks to their relatively lower cost, higher cycle of life and better security. Meanwhile, NMC batteries declined from 36% to 9% over the same period and NCA batteries dropped from 15% to less than 1% by 2024, as these chemistries have higher energy density than LFP. A small but growing share of other batteries[49] emerged in 2024 at 5%, reflecting the early stages of market entry for this alternative chemistry. The Na-ion category remained minimal throughout the years.

[49] *'Other' refers to technologies for long-duration energy storage, the commercialisation of which remains uncertain.*

Figure 9.4 Global market share trends of utility-scale battery technologies, 2021–2024



Based on: (BNEF, 2024c).

Advances in alternative battery technologies are progressing rapidly. China is leading in the development of sodium-ion and solid-state batteries for large-scale storage, thanks to significant investments in research and development (IEA, 2024).

UTILITY-SCALE BATTERY STORAGE APPLICATIONS

Since 2018, the primary use of electricity storage has been energy shifting (Figure 9.5), which accounts for 67% of the total capacity energy storage additions in 2024. Energy shifting balances the power system by storing renewable energy production at times of low market prices (leading to low/null revenue), or low demand (leading to curtailments). These circumstances encourage the storing of this energy for its later use at times of peak electricity demand or prices, resulting in improved system efficiency.

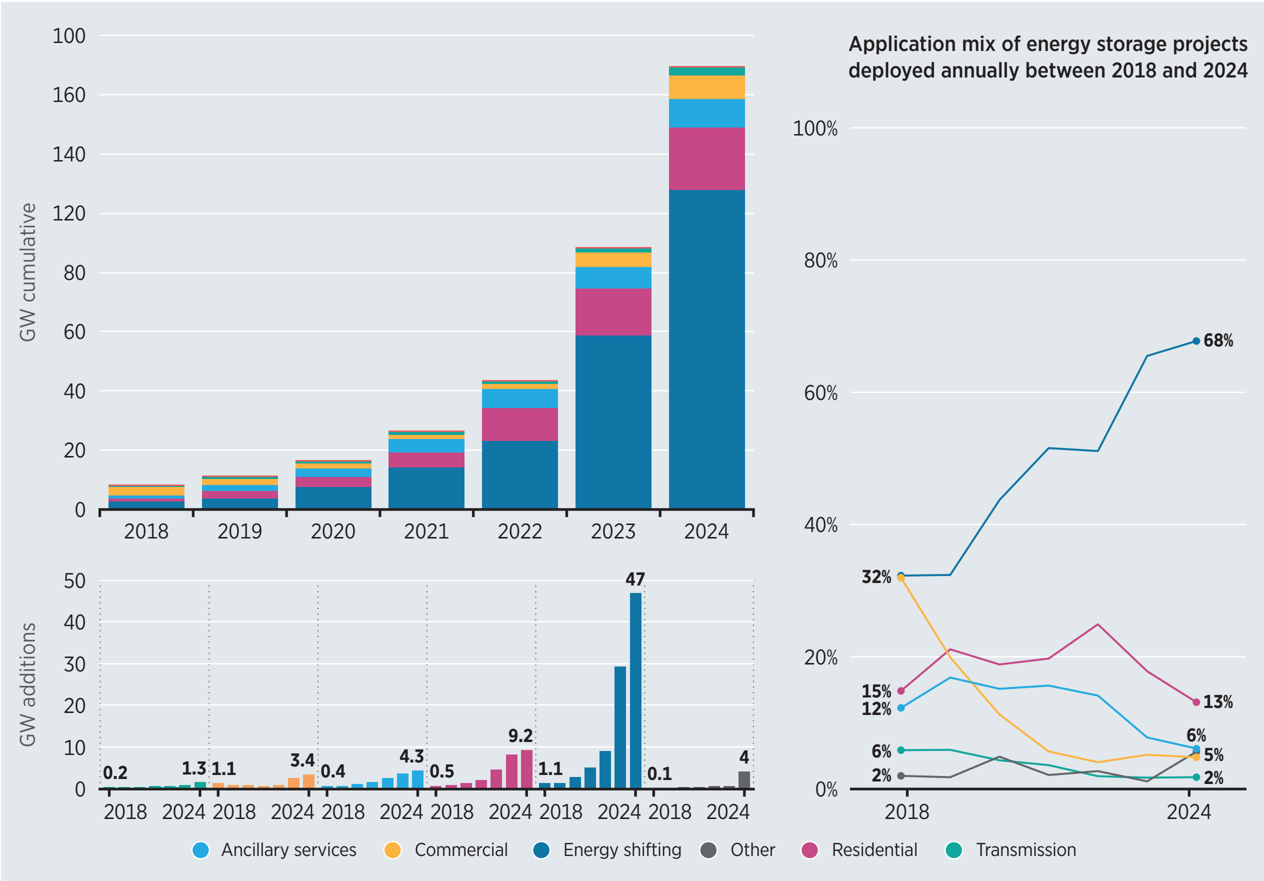
The increase in the use of BESS for energy shifting shows that falling electricity storage costs have led to economic opportunities for time shifting, especially as solar PV penetration has risen in certain markets. In the United States, solar PV with storage dominates the year-on-year growth of renewable hybrid projects, with this arrangement having the highest number of plants (Bolinger *et al.*, 2023), which are concentrated in California and Texas (Ember, 2025b). Additionally, hybrid power plants represent 55% of current installed solar capacity in the country (Gorman *et al.*, 2024).

BESS capacity for energy shifting added in 2024 was almost 47 GW; this value is 60% higher than the added capacity in 2023 and almost 70% of the total new BESS capacity additions in 2024. In both China and the United States, the large deployment of commissioned, utility-scale solar PV and wind projects – paired with energy storage – was the main driver of this increase in the share of capacity taken by energy shifting applications.[50]

[50] *These refer to the use of energy storage for renewable integration, price arbitrage and capacity services (BNEF, 2024a).*

In the residential segment, the capacity added reached 9 GW, but its share of the energy storage application mix fell from 18% in 2023 to 13% in 2024, as deployment in the utility-scale sector accelerated. Ancillary services also saw their share of the application mix decrease to 5% of all energy storage projects deployed annually in 2024. The new capacity added in that year accounted for 4.3 GW, compared to 3.5 GW in 2023. Transmission and distribution were marginal applications, despite the increase in year-on-year added capacity.

Figure 9.5 Battery storage deployment by application in GW



Based on: (BNEF, 2024c).
Note: GW = gigawatt.

HYBRID POWER SYSTEMS: INTEGRATING GENERATION AND STORAGE

Hybrid power plants combine different renewable generation technologies and/or integrate storage systems, creating flexible configurations that enhance energy reliability and system performance. By leveraging the complementary generation profiles of different technologies, these systems help align electricity supply with demand more effectively, mitigating issues such as intermittency and oversupply during peak generation periods. In addition, hybrid systems offer an effective response by maximising the use of available energy while alleviating constraints posed by limited grid connection points. The integration of battery storage, guided by system needs and supported by its inherent operational flexibility, makes these solutions increasingly vital for grid stability and the efficient integration of renewable energy sources (Intersolar, 2024).

With the growing deployment of hybrid systems, IRENA has been collecting data on these technologies to gain deeper insights into their cost optimisation. The Agency's database includes four different hybrid systems (solar[51] + BESS; wind + BESS; solar + wind; and solar + wind + BESS) with a combined total capacity of 10 GW (Figure 9.6) of projects with available cost data. Projects considered in this analysis were deployed in 2023 and 2024 across different countries, with the top markets in hybrid installations being the United States, China, India and Australia.

Figure 9.6 shows that solar + wind systems tend to have lower LCOE, as they do not include the additional cost of batteries. Their weighted average LCOE was USD 0.021/kWh, calculated based on a sample of ten projects deployed in 2023 and 2024 across different regions.

Figure 9.6 Characteristics of hybrid projects plants deployed in 2023 and 2024 by country

Notes: This analysis is based on a sample of projects with available cost data and does not represent the entire hybrid market; BESS = battery energy storage system; kWh = kilowatt hour; LCOE = levelised cost of electricity; USD = United States dollar.

Figure 9.7 shows hybrid project capacities, LCOE and capacity factors from the IRENA renewable costs database. The methodology applied for the LCOE calculation is described in Annex 1.

[51] *In all four hybrid systems, solar refers to solar photovoltaic.*

From the sample analysed, solar + BESS systems typically have the highest LCOE, with capacity factors generally ranging between 18% and 30%. The capacity factor of solar + wind + BESS systems varies widely, which largely depends on the specific mix and share of each technology. For wind + BESS systems, the available data is limited, making it difficult to draw any definitive conclusions.

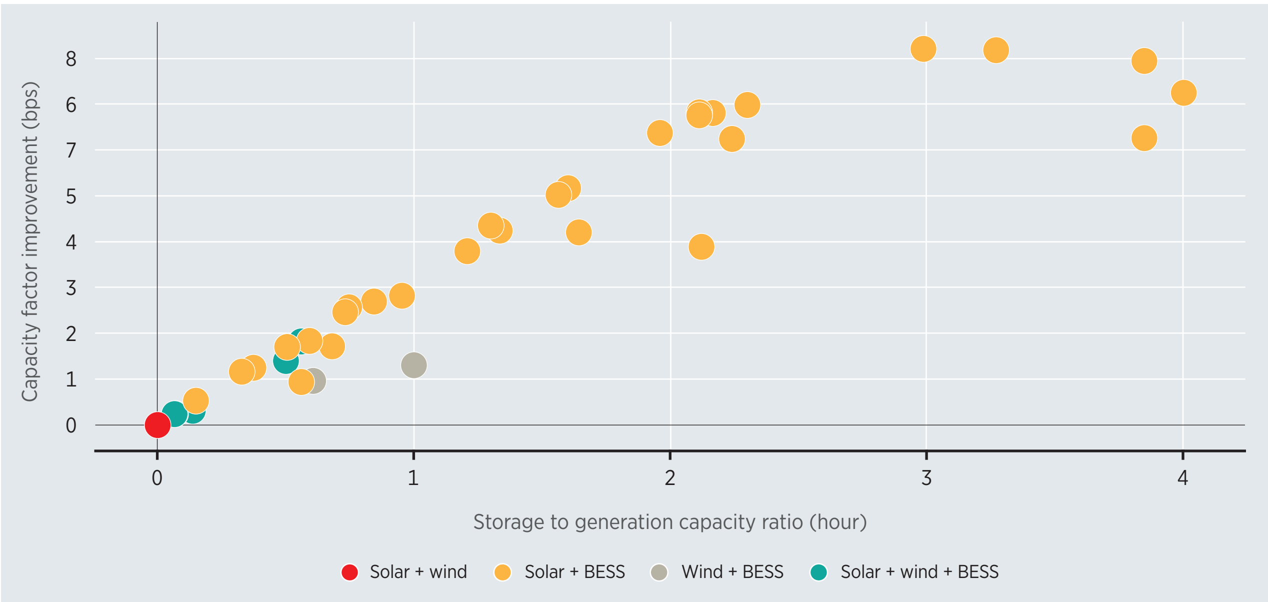
Hybrid systems typically achieve higher capacity factors than standalone solar or wind plants. This performance advantage is gained by leveraging complementary generation profiles and storage capabilities to deliver more consistent, efficient and higher capacity energy output.

When combining renewable technologies, hybrid systems can generate power more consistently, minimising periods of low or zero output and making better use of existing grid infrastructure. Storage adds another layer of flexibility, allowing excess energy to be saved and dispatched during times of high demand or low generation. This reduces curtailment and enhances the plant's ability to provide steady and dispatchable power (Murphy *et al.*, 2023).

Figure 9.7 shows the improvement in capacity factor of hybrid systems when compared to the storage and generation capacity ratio. Hybrid projects with a higher storage-to-generation capacity ratio have higher capacity factors.

Hybrid systems provide cost-saving advantages, including lower expenses for grid connections, land use, project development (such as feasibility studies), and operations and maintenance (O&M). They can also help reduce overall project financing costs by streamlining infrastructure and improving project efficiency. In hybrid systems, the LCOE for each renewable component is reduced due to shared investment and operational costs – particularly those related to grid interconnection and infrastructure. This cost efficiency makes hybrid projects more financially attractive compared to standalone systems (SolarPower Europe, 2025).

Figure 9.7 Capacity factor improvement vs. storage-to-generation capacity ratio



Notes: BESS = battery energy storage system; bps = basis points.